Code of Professional Ethics for Artificial Intelligence Developers in Reproductive Embryo Selection Systems

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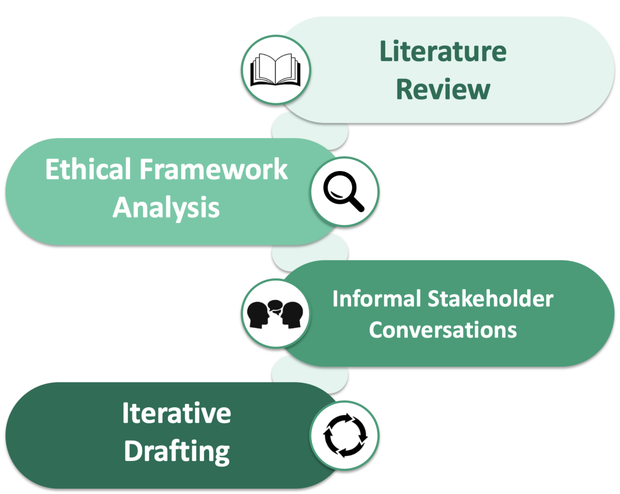
Introduction

- Artificial intelligence (AI) is being integrated into the fields of in vitro fertilization (IVF) and preimplantation genetic testing (PGT)
- Prospective parents can compare embryos for:
  - Disease risks *such as* type 2 diabetes, Alzheimer's disease, and hypertension
  - Social trait selection *such as* eye/hair color, height, and intelligence quotient (IQ)
- Although clinicians associated with these technologies act in accordance with professional oaths of service and bioethical principles, the AI developers who design and deploy these algorithms are not held to a comparable ethical framework

Objectives

1. Address the ethical accountability gap between clinicians and AI developers in IVF/PGT
2. Translate the principles of medical professionalism (e.g., beneficence, non-maleficence, primacy of patient welfare) into a practical and professional Code of Ethics for AI developers
3. Incorporate stakeholder input from AI developers, embryologists, and bioethicists

Methods

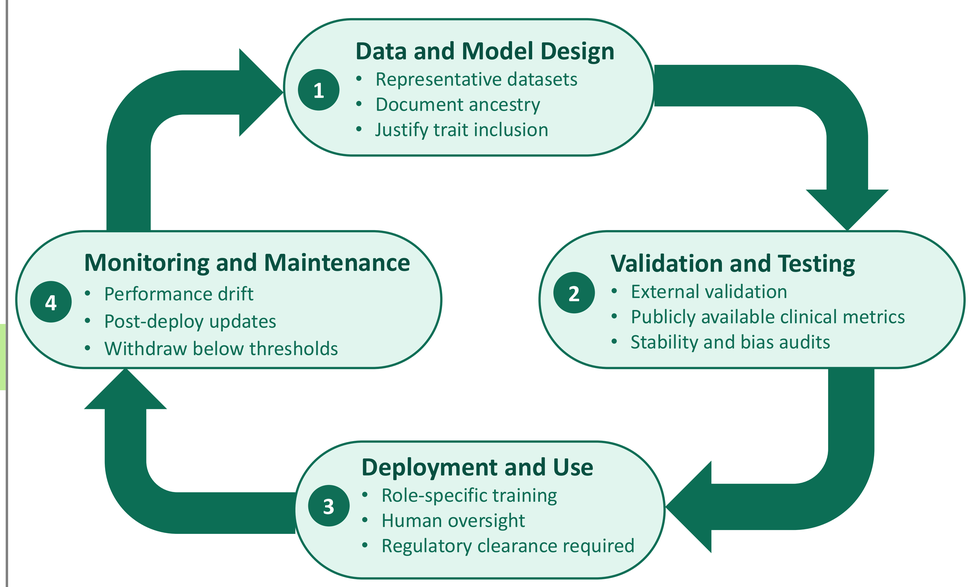


Results

Six Foundational Professional Commitments:

- Responsible Embryo-Selection System Design
- Transparent Presentation of Embryo Risk Scores
- Support for Informed Reproductive Counseling
- Equity in Polygenic Prediction
- Scientific and Technical Integrity
- Independence from Market Pressures

Obligations Across the AI Development Lifecycle:



Prohibited Practices:

- No social trait prioritization
- No unvalidated deployment
- No deterministic claims
- No concealment of flaws

Limitations

- Stakeholder conversations were conducted informally
- This Code has not yet been empirically validated through formal stakeholder review (e.g., Delphi survey) or tested for adoption in industry or professional society settings

Conclusions and Next Steps

- The resulting framework found that explicit ethical standards were necessary to ensure that AI developers safeguard trust with prospective parents, future children, and the general public.
- In the future, this project will involve stakeholder review and the exploration of potential pathways for adopting this Code through professional societies and fertility clinics.

Acknowledgements

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References

Access the references page using the QR code on the right:

Full Code of Ethics

Access the full Code of Ethics using the QR code on the right. The website also invites structured feedback, input that will inform future revisions of this Code.